Surface Metrics Analysis

Analysis of TextDescriptives Metrics (Readability, Complexity, POS)

PENPAL Project

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0.1 Summary

- AI text is more syntactically complex and less readable: longer dependency distances, higher grade-level estimates, more nouns/adjectives/determiners.
- Human text is more variable and conversational, using more pronouns, auxiliaries, adverbs, interjections, and shorter dependency structures.
- Several linguistic features change over turn count (e.g., punctuation, entropy, document length), with differing slopes for user vs AI.
- The two agents maintain distinct stylistic signatures across the interaction, not converging linguistically.
- Overall, AI emphasises structural complexity, while humans contribute stylistic diversity and natural conversational markers.

1 1. Data Loading and Preprocessing

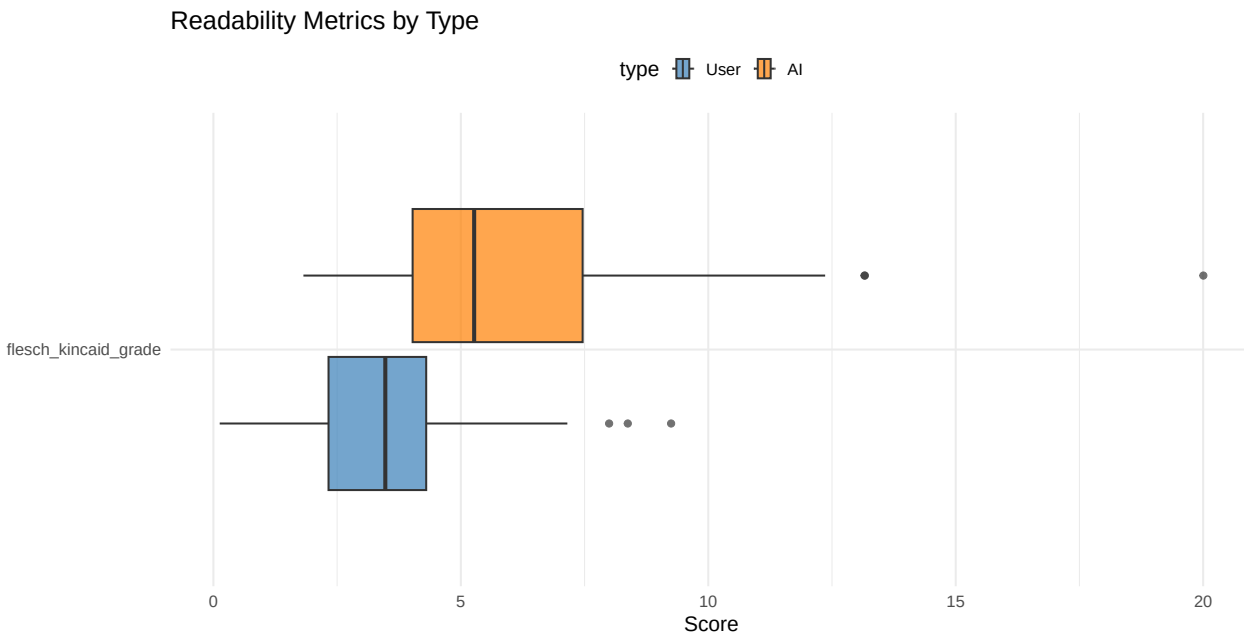
1.1 1.1 Load Data

```
## Full Story Data Dimensions: 174 69
```

```
## Interaction Data Dimensions: 1566 72
```

2 2. Full Story Analysis

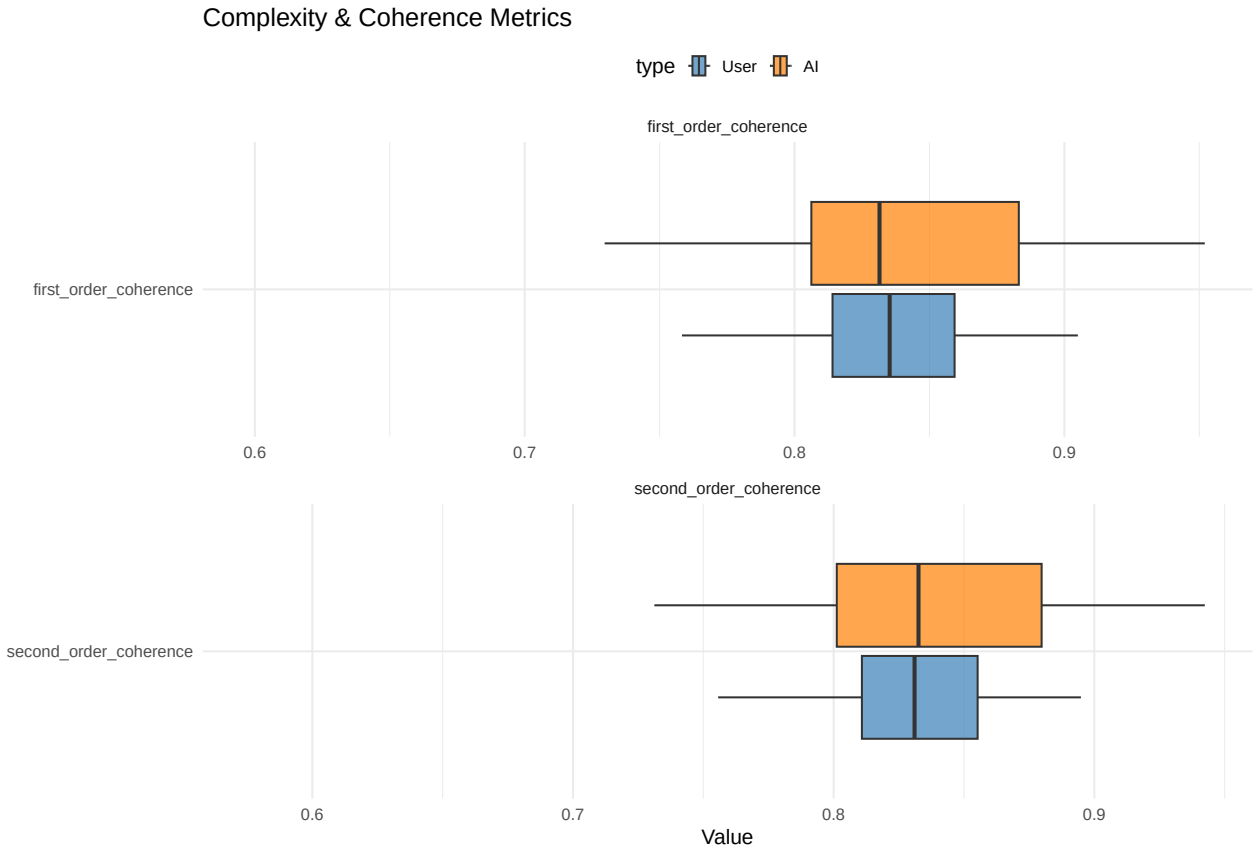
2.1 2.1 Readability Metrics



2.1.1 Statistical Models (Readability)

```
##
## === Model: flesch_kincaid_grade ===
##              Estimate Std. Error       df   t value     Pr(>|t|)
## (Intercept) 3.567648  0.2659087 156.5283 13.416812 8.300750e-28
## typeAI      2.371539  0.3113762  86.0000  7.616315 3.137731e-11
```

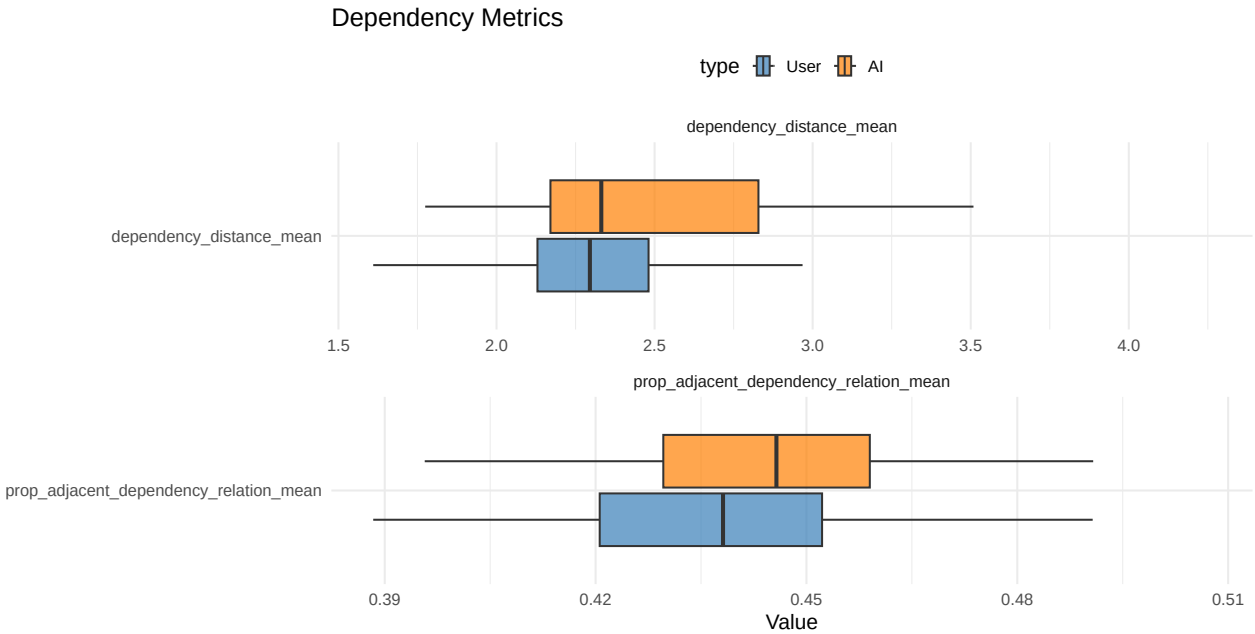
2.2 2.2 Complexity & Coherence



2.2.1 Statistical Models (Complexity & Coherence)

```
##
## === Model: first_order_coherence ===
##                Estimate  Std. Error       df   t value      Pr(>|t|)
## (Intercept) 0.831514092 0.005434711 158.6276 153.00061 3.448810e-174
## typeAI      0.006520275 0.006474628  86.0000   1.00705  3.167358e-01
##
## === Model: second_order_coherence ===
##                Estimate  Std. Error       df    t value      Pr(>|t|)
## (Intercept) 0.828257508 0.005374625 158.9121 154.105168 6.260119e-175
## typeAI      0.007579357 0.006418197  86.0000   1.180917  2.408915e-01
```

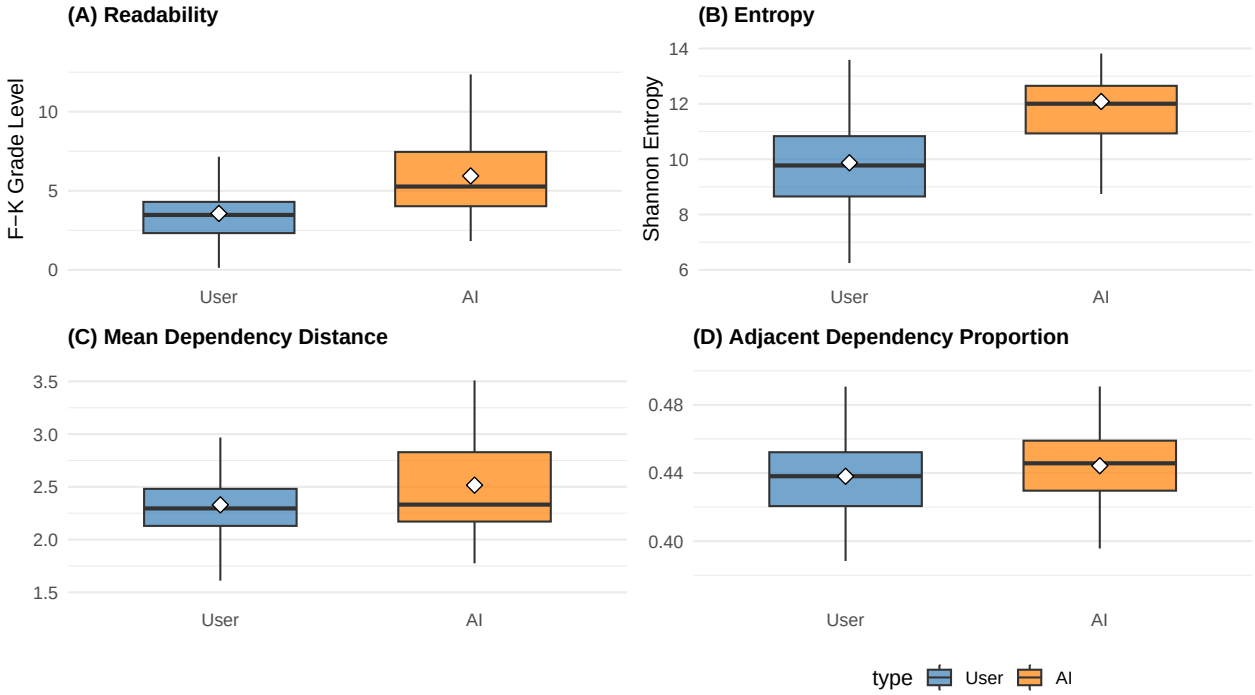
2.3 2.3 Dependency Metrics



2.3.1 Statistical Models (Dependency)

```
##
## === Model: dependency_distance_mean ===
##              Estimate Std. Error       df   t value      Pr(>|t|)
## (Intercept) 2.3290103 0.04446139 167.9721 52.382765 5.590885e-106
## typeAI      0.1867605 0.05780481  86.0000  3.230882  1.749369e-03
##
## === Model: prop_adjacent_dependency_relation_mean ===
##                Estimate  Std. Error       df    t value      Pr(>|t|)
## (Intercept) 0.438083467 0.002420165 166.4342 181.013911 4.937504e-193
## typeAI      0.006200531 0.003093895  86.0000   2.004118  4.820427e-02
```

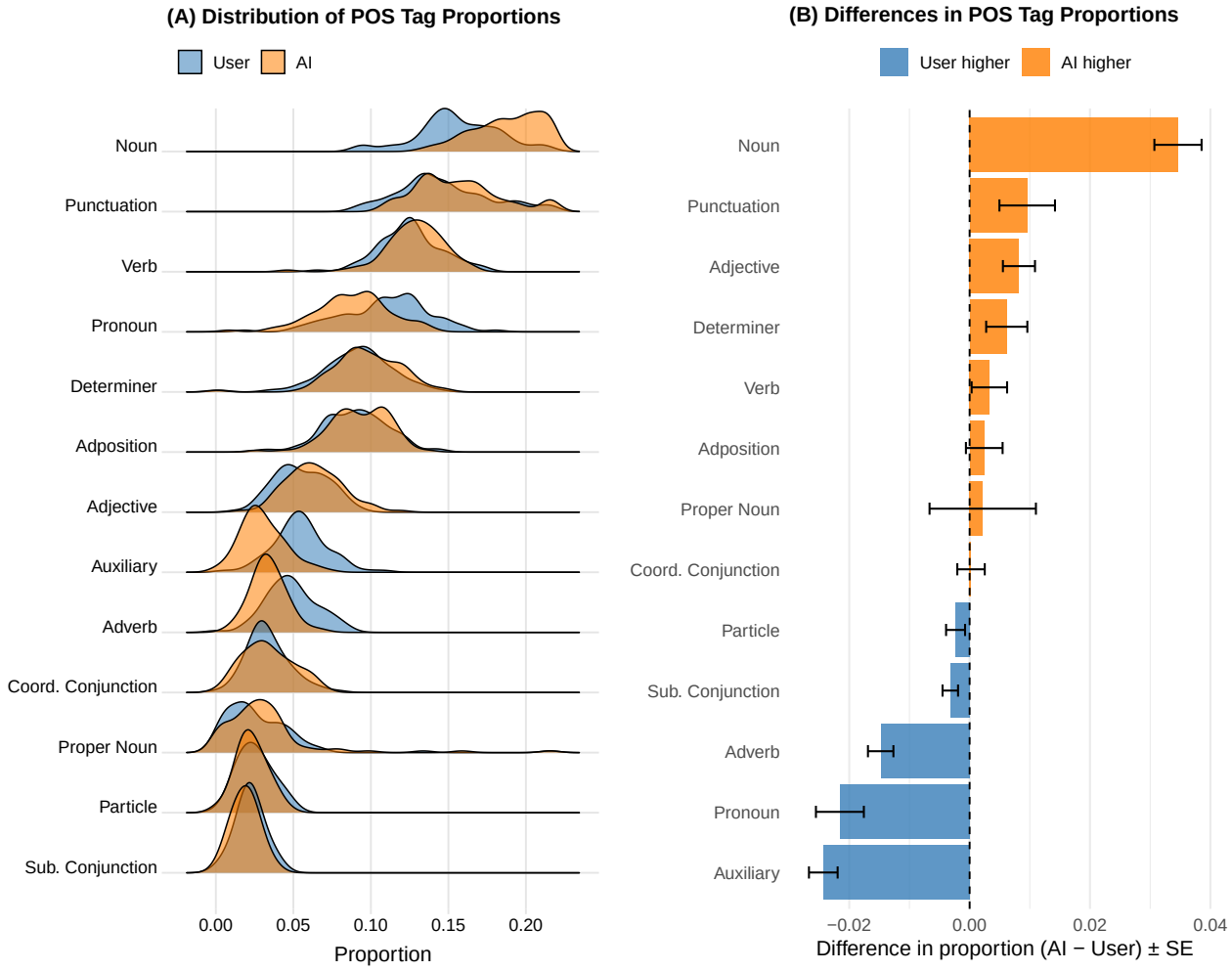
2.4 Combined Plots

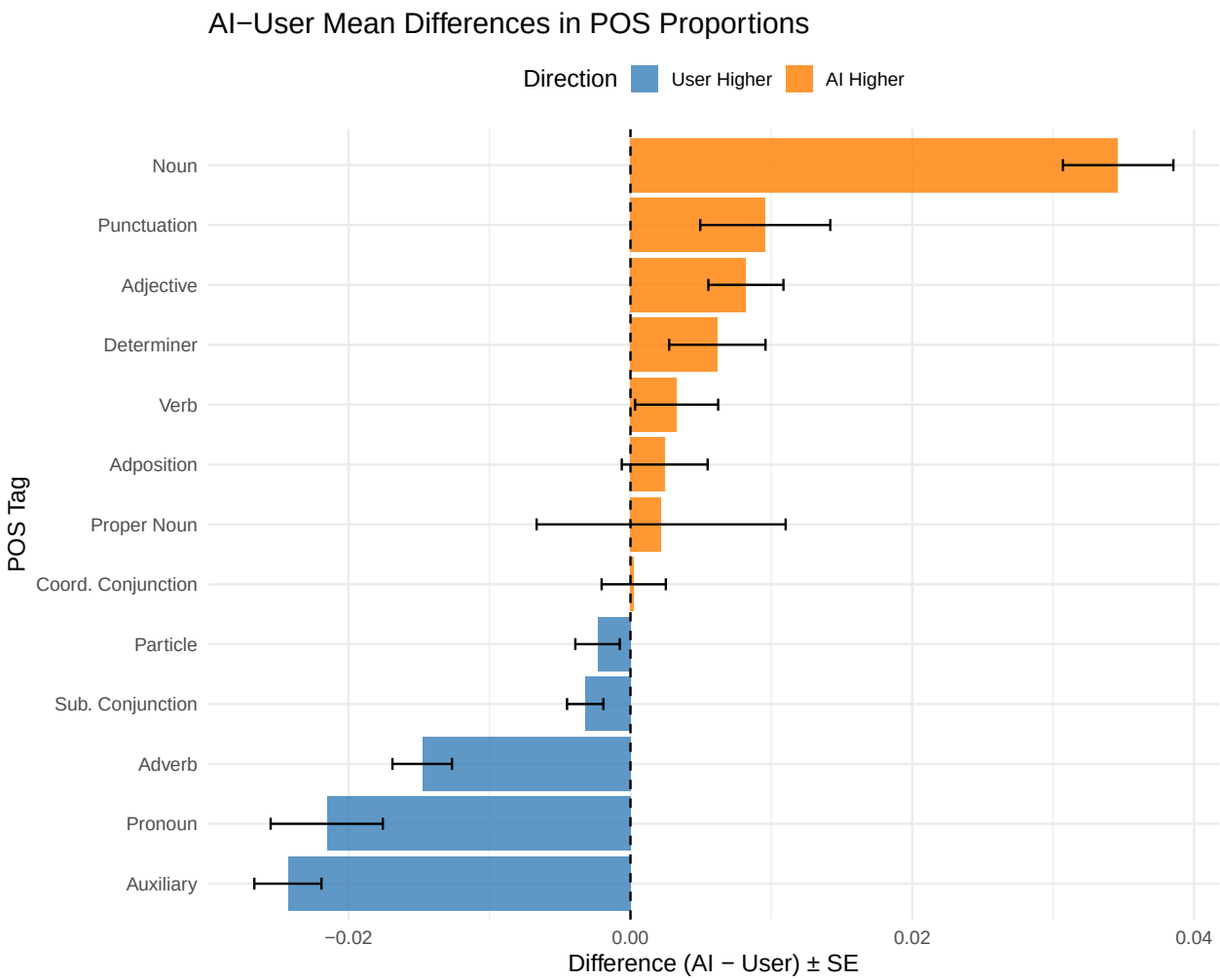


Note: White diamonds indicate means. Boxes show median and IQR.

2.5 2.4 Part-of-Speech (POS) Analysis

2.5.1 AI-User Mean Differences





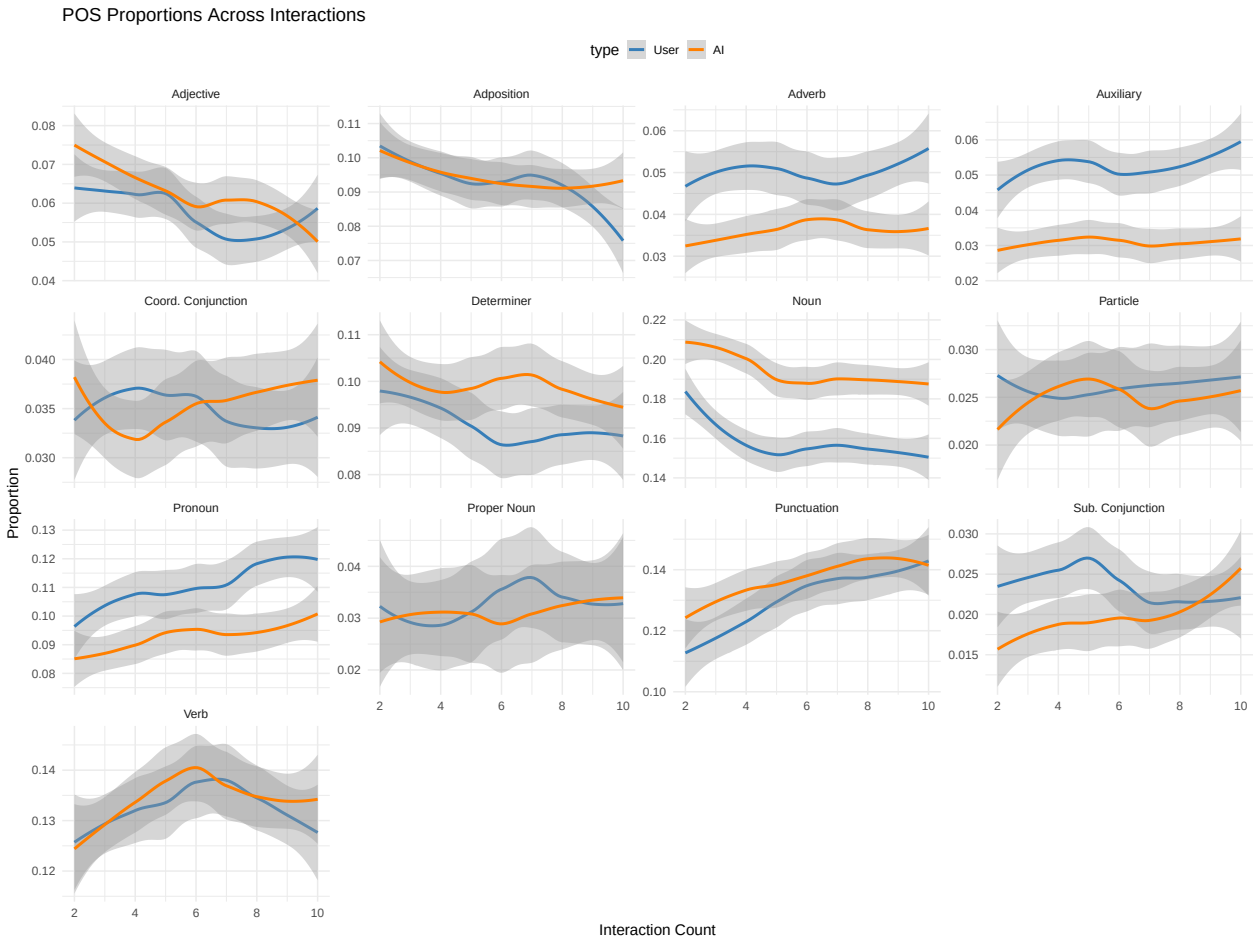
2.5.2 Statistical Models (POS)

```
##
## === POS Model: Adjective ===
##                Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept) 0.055158309 0.001883902 164.3082 29.278752 4.299099e-67
## typeAI      0.008194314 0.002358472  86.0000  3.474416 8.042693e-04
##
## === POS Model: Adposition ===
##                Estimate  Std. Error      df   t value     Pr(>|t|)
## (Intercept) 0.090281272 0.002154472 151.771 41.904121 2.543210e-85
## typeAI      0.002430637 0.002427807  86.000  1.001166 3.195553e-01
##
## === POS Model: Adverb ===
##                Estimate  Std. Error       df  t value     Pr(>|t|)
## (Intercept)  0.04873043 0.001494801 161.8285 32.59994 5.115000e-73
## typeAI      -0.01477759 0.001829888  86.0000 -8.07568 3.718941e-12
##
## === POS Model: Auxiliary ===
##                Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept)  0.05453863 0.001683831 159.6497  32.38961 4.476342e-72
## typeAI      -0.02431195 0.002023214  86.0000 -12.01650 4.255855e-20
```

```
##
## === POS Model: Coord. Conjunction ===
##                 Estimate  Std. Error       df    t value     Pr(>|t|)
## (Intercept) 0.0335449145 0.001611159 165.3515 20.8203615 4.504239e-48
## typeAI      0.0002328712 0.002037311  86.0000  0.1143032 9.092639e-01
##
## === POS Model: Determiner ===
##                Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept) 0.091416884 0.002418079 126.4238 37.805589 9.149427e-71
## typeAI      0.006165521 0.002161663  86.0000  2.852213 5.436210e-03
##
## === POS Model: Noun ===
##               Estimate  Std. Error       df  t value     Pr(>|t|)
## (Intercept) 0.15577240 0.002771479 135.9525 56.20552 5.399914e-96
## typeAI      0.03461315 0.002729802  86.0000 12.67973 2.179267e-21
##
## === POS Model: Particle ===
##                 Estimate Std. Error       df   t value     Pr(>|t|)
## (Intercept)  0.025397358 0.00111547 166.7699 22.768301 4.769644e-53
## typeAI      -0.002335403 0.00143103  86.0000 -1.631973 1.063417e-01
##
## === POS Model: Pronoun ===
##                Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept)  0.10974262 0.002814483 135.3225 38.992105 1.838279e-75
## typeAI      -0.02155423 0.002755858  86.0000 -7.821243 1.214363e-11
##
## === POS Model: Proper Noun ===
##               Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept) 0.03223026 0.006248714 93.38988 5.1579039 1.395762e-06
## typeAI      0.00217675 0.002538783 86.00000 0.8573987 3.936076e-01
##
## === POS Model: Punctuation ===
##                Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept) 0.147239592 0.003263083 134.8114 45.122840 3.143397e-83
## typeAI      0.009566825 0.003179722  86.0000  3.008699 3.441259e-03
##
## === POS Model: Sub. Conjunction ===
##                 Estimate   Std. Error        df   t value     Pr(>|t|)
## (Intercept)  0.022113982 0.0009106799 170.73259 24.282936 2.888317e-57
## typeAI      -0.003211366 0.0012311644  85.99999 -2.608397 1.072522e-02
##
## === POS Model: Verb ===
##                Estimate  Std. Error       df   t value      Pr(>|t|)
## (Intercept) 0.125181648 0.002084527 161.5986 60.052790 1.995227e-112
## typeAI      0.003276804 0.002546702  86.0000  1.286685  2.016568e-01
```

3 3. Interaction Level Analysis

3.1 3.1 POS Proportions Over Time



3.1.1 Statistical Models (POS Interaction)

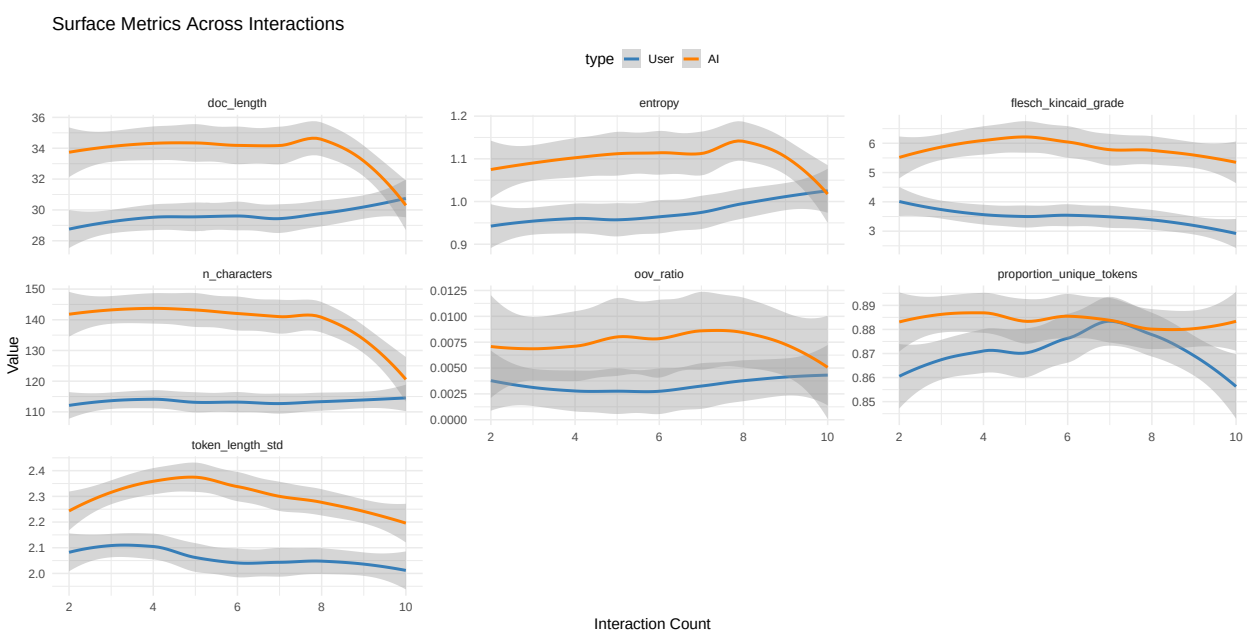
```
##
## === POS Interaction Model: Adjective ===
##                              Estimate   Std. Error       df   t value
## (Intercept)               0.065586862 0.0039648986 1297.390 16.541876
## interaction_count        -0.001319758 0.0005801336 1475.047 -2.274921
## typeAI                    0.012513671 0.0053649008 1475.212  2.332507
## interaction_count:typeAI -0.001258426 0.0008210837 1475.167 -1.532641
##                              Pr(>|t|)
## (Intercept)              6.438713e-56
## interaction_count        2.305368e-02
## typeAI                   1.980793e-02
## interaction_count:typeAI 1.255788e-01
##
## === POS Interaction Model: Adposition ===
##                              Estimate   Std. Error       df   t value
## (Intercept)               0.107966259 0.0042230664 1096.499 25.565845
## interaction_count        -0.002578829 0.0006009338 1475.001 -4.291369
## typeAI                   -0.006674207 0.0055572961 1475.124 -1.200981
```

```
## interaction_count:typeAI  0.001443501 0.0008505275 1475.090  1.697183
##                              Pr(>|t|)
## (Intercept)              1.857406e-113
## interaction_count         1.891400e-05
## typeAI                    2.299514e-01
## interaction_count:typeAI  8.987296e-02
##
## === POS Interaction Model: Adverb ===
##                              Estimate   Std. Error       df   t value
## (Intercept)               0.0468799412 0.0035120467 1403.388 13.348325
## interaction_count         0.0005845294 0.0005218621 1475.062  1.120084
## typeAI                   -0.0134737170 0.0048259915 1475.264 -2.791906
## interaction_count:typeAI -0.0001758888 0.0007386064 1475.209 -0.238136
##                             Pr(>|t|)
## (Intercept)              2.373577e-38
## interaction_count        2.628603e-01
## typeAI                   5.307435e-03
## interaction_count:typeAI 8.118087e-01
##
## === POS Interaction Model: Auxiliary ===
##                              Estimate   Std. Error       df   t value
## (Intercept)               0.0467201494 0.0034292633 1253.262 13.623961
## interaction_count         0.0009958628 0.0004987174 1474.998  1.996848
## typeAI                   -0.0172059305 0.0046119974 1475.152 -3.730690
## interaction_count:typeAI -0.0007735075 0.0007058535 1475.110 -1.095847
##                             Pr(>|t|)
## (Intercept)              1.619151e-39
## interaction_count        4.602520e-02
## typeAI                   1.981965e-04
## interaction_count:typeAI 2.733246e-01
##
## === POS Interaction Model: Coord. Conjunction ===
##                              Estimate   Std. Error        df   t value
## (Intercept)               0.0368793112 0.0028276491  979.0172 13.042393
## interaction_count        -0.0003317172 0.0003953905 1474.9744 -0.838961
## typeAI                   -0.0040855699 0.0036564911 1475.0798 -1.117347
## interaction_count:typeAI  0.0007746788 0.0005596145 1475.0510  1.384308
##                             Pr(>|t|)
## (Intercept)              5.766757e-36
## interaction_count        4.016271e-01
## typeAI                   2.640279e-01
## interaction_count:typeAI 1.664736e-01
##
## === POS Interaction Model: Determiner ===
##                              Estimate   Std. Error        df    t value
## (Intercept)               0.0985968980 0.0043850604  871.1129 22.4847297
## interaction_count        -0.0012397912 0.0006022332 1475.0114 -2.0586563
## typeAI                    0.0050134445 0.0055693438 1475.1027  0.9001859
## interaction_count:typeAI  0.0004335937 0.0008523701 1475.0777  0.5086919
##                             Pr(>|t|)
## (Intercept)              1.197943e-88
## interaction_count        3.970246e-02
## typeAI                   3.681683e-01
## interaction_count:typeAI 6.110443e-01
```

```
##
## === POS Interaction Model: Noun ===
##                              Estimate   Std. Error       df     t value
## (Intercept)               1.760079e-01 0.0053924736 1022.704 32.63954924
## interaction_count        -2.911488e-03 0.0007592071 1474.765 -3.83490680
## typeAI                    3.585451e-02 0.0070209845 1474.877  5.10676457
## interaction_count:typeAI  7.338512e-05 0.0010745401 1474.846  0.06829445
##                              Pr(>|t|)
## (Intercept)              1.068572e-160
## interaction_count         1.309050e-04
## typeAI                    3.704585e-07
## interaction_count:typeAI  9.455605e-01
##
## === POS Interaction Model: Particle ===
##                              Estimate   Std. Error       df    t value
## (Intercept)               2.534686e-02 0.0025914771 1400.173  9.7808532
## interaction_count         1.321472e-04 0.0003848893 1474.984  0.3433383
## typeAI                   -1.730170e-03 0.0035593174 1475.185 -0.4860961
## interaction_count:typeAI  9.076896e-05 0.0005447450 1475.130  0.1666265
##                             Pr(>|t|)
## (Intercept)              6.712917e-22
## interaction_count        7.313929e-01
## typeAI                   6.269712e-01
## interaction_count:typeAI 8.676868e-01
##
## === POS Interaction Model: Pronoun ===
##                             Estimate   Std. Error        df   t value
## (Intercept)               0.093953312 0.0050477952  885.1239 18.612743
## interaction_count         0.002843734 0.0006949623 1475.0144  4.091925
## typeAI                   -0.011568005 0.0064268841 1475.1074 -1.799940
## interaction_count:typeAI -0.001112414 0.0009836140 1475.0820 -1.130946
##                             Pr(>|t|)
## (Intercept)              1.646685e-65
## interaction_count        4.509534e-05
## typeAI                   7.207439e-02
## interaction_count:typeAI 2.582616e-01
##
## === POS Interaction Model: Proper Noun ===
##                              Estimate   Std. Error        df    t value
## (Intercept)               0.0298190482 0.0067364829  124.3923  4.4265010
## interaction_count         0.0004203809 0.0004422566 1475.0018  0.9505361
## typeAI                   -0.0015749511 0.0040899690 1475.0085 -0.3850765
## interaction_count:typeAI  0.0000945596 0.0006259541 1475.0066  0.1510647
##                             Pr(>|t|)
## (Intercept)              2.070126e-05
## interaction_count        3.419957e-01
## typeAI                   7.002362e-01
## interaction_count:typeAI 8.799453e-01
##
## === POS Interaction Model: Punctuation ===
##                             Estimate   Std. Error        df   t value
## (Intercept)               0.108104991 0.0051188024  620.0101 21.119196
## interaction_count         0.003694854 0.0006649991 1475.0100  5.556180
## typeAI                    0.014858934 0.0061498210 1475.0733  2.416157
```

```
## interaction_count:typeAI -0.001385053 0.0009412090 1475.0560 -1.471568
##                              Pr(>|t|)
## (Intercept)              5.341295e-75
## interaction_count        3.266870e-08
## typeAI                   1.580608e-02
## interaction_count:typeAI 1.413509e-01
##
## === POS Interaction Model: Sub. Conjunction ===
##                              Estimate   Std. Error       df   t value
## (Intercept)               0.0258539544 0.0022703561 1493.712 11.387621
## interaction_count        -0.0004014938 0.0003427612 1475.062 -1.171351
## typeAI                   -0.0117281727 0.0031696977 1475.322 -3.700092
## interaction_count:typeAI  0.0013536597 0.0004851160 1475.251  2.790383
##                              Pr(>|t|)
## (Intercept)              7.193250e-29
## interaction_count        2.416467e-01
## typeAI                   2.234394e-04
## interaction_count:typeAI 5.332337e-03
##
## === POS Interaction Model: Verb ===
##                              Estimate   Std. Error       df    t value
## (Intercept)               0.1304780968 0.0043183375 1294.097 30.2148911
## interaction_count         0.0002669916 0.0006315978 1474.881  0.4227241
## typeAI                   -0.0022844396 0.0058408263 1475.046 -0.3911158
## interaction_count:typeAI  0.0006554429 0.0008939228 1475.001  0.7332210
##                              Pr(>|t|)
## (Intercept)              3.338825e-152
## interaction_count         6.725582e-01
## typeAI                    6.957681e-01
## interaction_count:typeAI  4.635402e-01
```

3.2 3.2 Other Metrics Over Time



3.2.1 Statistical Models (Other Metrics Interaction)

```
##
## === Metric Interaction Model: flesch_kincaid_grade ===
##                            Estimate Std. Error        df   t value     Pr(>|t|)
## (Intercept)               4.1108921 0.30333304  401.0121 13.552405 1.033290e-34
## interaction_count        -0.1050781 0.03609219 1475.0102 -2.911381 3.652409e-03
## typeAI                    1.9341885 0.33377697 1475.0508  5.794853 8.346939e-09
## interaction_count:typeAI  0.0657951 0.05108337 1475.0397  1.287995 1.979499e-01
##
## === Metric Interaction Model: doc_length ===
##                            Estimate Std. Error       df   t value      Pr(>|t|)
## (Intercept)              28.5375479 0.72901189  257.492 39.145518 2.080713e-110
## interaction_count         0.1875479 0.07636032 1476.000  2.456091  1.416028e-02
## typeAI                    6.6731801 0.70538645 1476.000  9.460318  1.164874e-20
## interaction_count:typeAI -0.4327586 0.10798981 1476.000 -4.007403  6.445175e-05
##
## === Metric Interaction Model: entropy ===
##                              Estimate  Std. Error        df   t value
## (Intercept)               0.918255372 0.029837992  386.1319 30.774704
## interaction_count         0.009836739 0.003518403 1476.0000  2.795797
## typeAI                    0.185607893 0.032501610 1476.0000  5.710729
## interaction_count:typeAI -0.010546669 0.004975773 1476.0000 -2.119604
##                               Pr(>|t|)
## (Intercept)              6.038068e-106
## interaction_count         5.244272e-03
## typeAI                    1.358047e-08
## interaction_count:typeAI  3.420596e-02
##
## === Metric Interaction Model: n_characters ===
##                             Estimate Std. Error        df    t value
## (Intercept)              112.7459770  3.0558487  256.5859 36.8951441
## interaction_count          0.1266284  0.3196824 1476.0000  0.3961068
## typeAI                    38.0717752  2.9530993 1476.0000 12.8921417
## interaction_count:typeAI  -2.0647510  0.4520992 1476.0000 -4.5670312
##                               Pr(>|t|)
## (Intercept)              1.373845e-104
## interaction_count         6.920835e-01
## typeAI                    4.092807e-36
## interaction_count:typeAI  5.358243e-06
##
## === Metric Interaction Model: oov_ratio ===
##                               Estimate   Std. Error        df    t value
## (Intercept)               0.0025963677 0.0021341941  152.3204  1.2165565
## interaction_count         0.0001366069 0.0001705034 1476.0000  0.8011973
## typeAI                    0.0049271405 0.0015750432 1476.0000  3.1282574
## interaction_count:typeAI -0.0001661326 0.0002411283 1476.0000 -0.6889803
##                             Pr(>|t|)
## (Intercept)              0.225654733
## interaction_count        0.423146410
## typeAI                   0.001792843
## interaction_count:typeAI 0.490943981
##
## === Metric Interaction Model: proportion_unique_tokens ===
```

```
##                              Estimate   Std. Error       df      t value
## (Intercept)               8.702172e-01 0.0060996077 1100.274 142.66772948
## interaction_count         3.006961e-05 0.0008684659 1474.928   0.03462382
## typeAI                    1.748715e-02 0.0080313702 1475.051   2.17735548
## interaction_count:typeAI -6.998619e-04 0.0012291772 1475.017  -0.56937428
##                            Pr(>|t|)
## (Intercept)              0.00000000
## interaction_count        0.97238439
## typeAI                   0.02961211
## interaction_count:typeAI 0.56918886
##
## === Metric Interaction Model: token_length_std ===
##                              Estimate  Std. Error        df     t value
## (Intercept)               2.1287402059 0.036056407  728.9237 59.03916569
## interaction_count        -0.0109814362 0.004814068 1475.0069 -2.28111350
## typeAI                    0.2288189390 0.044519760 1475.0818  5.13971637
## interaction_count:typeAI  0.0004696355 0.006813600 1475.0613  0.06892618
##                              Pr(>|t|)
## (Intercept)              5.841813e-280
## interaction_count         2.268375e-02
## typeAI                    3.119843e-07
## interaction_count:typeAI  9.450577e-01
```

4 4. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] broom.mixed_0.2.9.6 cowplot_1.2.0       patchwork_1.3.2
##  [4] here_1.0.2          arrow_22.0.0.1      lmerTest_3.1-3
##  [7] lme4_1.1-38         Matrix_1.7-1        lubridate_1.9.4
## [10] forcats_1.0.1       stringr_1.6.0       dplyr_1.1.4
## [13] purrr_1.2.0         readr_2.1.6         tidyr_1.3.1
## [16] tibble_3.3.0        ggplot2_4.0.1       tidyverse_2.0.0
## [19] ggridges_0.5.7
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.55           lattice_0.22-6
```

```
##  [4] tzdb_0.5.0          numDeriv_2016.8-1.1 vctrs_0.6.5
##  [7] tools_4.4.2         Rdpack_2.6.4        generics_0.1.4
## [10] parallel_4.4.2      pacman_0.5.1        pkgconfig_2.0.3
## [13] RColorBrewer_1.1-3  S7_0.2.1            assertthat_0.2.1
## [16] lifecycle_1.0.4     compiler_4.4.2      farver_2.1.2
## [19] textshaping_1.0.4   tinytex_0.58        codetools_0.2-20
## [22] htmltools_0.5.9     yaml_2.3.12         pillar_1.11.1
## [25] furrr_0.3.1         nloptr_2.2.1        MASS_7.3-61
## [28] reformulas_0.4.2    boot_1.3-31         parallelly_1.45.1
## [31] nlme_3.1-166        tidyselect_1.2.1    digest_0.6.39
## [34] future_1.68.0       stringi_1.8.7       listenv_0.10.0
## [37] labeling_0.4.3      splines_4.4.2       rprojroot_2.1.1
## [40] fastmap_1.2.0       grid_4.4.2          cli_3.6.5
## [43] magrittr_2.0.4      broom_1.0.11        withr_3.0.2
## [46] scales_1.4.0        backports_1.5.0     bit64_4.6.0-1
## [49] timechange_0.3.0    rmarkdown_2.30      globals_0.18.0
## [52] bit_4.6.0           ragg_1.5.0          hms_1.1.4
## [55] evaluate_1.0.5      knitr_1.51          rbibutils_2.4
## [58] mgcv_1.9-1          rlang_1.1.6         Rcpp_1.1.0
## [61] glue_1.8.0          rstudioapi_0.17.1   minqa_1.2.8
## [64] R6_2.6.1            systemfonts_1.3.1
```